

## Unit 113 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 73.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 36 plots, 411 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

**Table 1. Stand summary from the BAF 20 cruise, per acre**

| STAND DENSITY        |                                    | STAND INFO          |                                                |
|----------------------|------------------------------------|---------------------|------------------------------------------------|
| Basal area           | 228 sq ft/ac (SE 2.0 %)            | Type                | Sierran mixed conifer                          |
| Trees per acre       | 70                                 | CWHR size / density | 5 / D (cover 95 %)                             |
| QMD                  | 24.5 in                            | SDI                 | 283 (49 % of max 573)                          |
| SAWTIMBER (>= 10 in) |                                    | BIOMASS (< 10 in)   |                                                |
| BA   QMD             | 228 sq ft/ac   24.5 in             | BA   QMD            | 0 sq ft/ac   0.0 in                            |
| TPA                  | 70                                 | TPA                 | 0                                              |
| Volume               | 39.7 MBF/ac (7904 cu ft)           | Volume              | 0.0 green tons/ac (0 cu ft)                    |
| TREATMENT TARGET     |                                    | CRUISE DESIGN       |                                                |
| Leave                | 35 % of max SDI = ~162 sq ft/ac BA | Sampling error      | 4.1 % at 95 % (t = 2.03, n = 36)               |
| Species (BA)         | PP 70, WF 66, IC 36, DF 34, SP 23  | Stratum standard    | 40 % - MEETS; 36 plots vs 20-plot practice: ok |

Cubic volume = total-stem CVTS by species (PNW-FIA California equations, MacLean and Berger 1976, CARB 2011 compendium) net of defect; Scribner board feet at 5.02 bf/cu ft (Keegan et al. 2010, table 2, California 2000-2006); green tons at PP 45, WF 47, DF 38, SP 49, IC 45 lb/cu ft of wood (Miles and Smith 2009, NRS-38 table 1A). SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum (table 3.5.1, rev. 2025-09-23). Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1). Sources and URLs: cruise\_data.xlsx, Standards and assumptions.

**Table 2. Stand table by 4-inch DBH class, per acre**

| DBH class, in | Trees      | TPA         | BA           | cu ft       | MBF         |
|---------------|------------|-------------|--------------|-------------|-------------|
| 8-11.9        | 3          | 2.7         | 1.7          | 42          | 0.2         |
| 12-15.9       | 18         | 9.2         | 10.0         | 259         | 1.3         |
| 16-19.9       | 45         | 13.9        | 25.0         | 676         | 3.4         |
| 20-23.9       | 81         | 17.3        | 45.0         | 1363        | 6.8         |
| 24-27.9       | 70         | 10.6        | 38.9         | 1290        | 6.5         |
| 28-31.9       | 50         | 5.8         | 27.8         | 943         | 4.7         |
| 32-35.9       | 49         | 4.3         | 27.2         | 1039        | 5.2         |
| 36-39.9       | 43         | 3.1         | 23.9         | 999         | 5.0         |
| 40-43.9       | 21         | 1.2         | 11.7         | 500         | 2.5         |
| 44-47.9       | 13         | 0.6         | 7.2          | 311         | 1.6         |
| 48-51.9       | 5          | 0.2         | 2.8          | 127         | 0.6         |
| 52-55.9       | 5          | 0.2         | 2.8          | 127         | 0.6         |
| 56-59.9       | 4          | 0.1         | 2.2          | 107         | 0.5         |
| 60-63.9       | 4          | 0.1         | 2.2          | 120         | 0.6         |
| <b>All</b>    | <b>411</b> | <b>69.5</b> | <b>228.3</b> | <b>7904</b> | <b>39.7</b> |

**Table 3. Stock table by species, per acre**

| Species    | Trees      | TPA         | BA           | QMD         | cu ft       | MBF         |
|------------|------------|-------------|--------------|-------------|-------------|-------------|
| PP         | 126        | 20.3        | 70.0         | 25.2        | 2704        | 13.6        |
| WF         | 118        | 19.6        | 65.6         | 24.8        | 2300        | 11.5        |
| IC         | 64         | 10.7        | 35.6         | 24.6        | 999         | 5.0         |
| DF         | 62         | 12.7        | 34.4         | 22.3        | 1093        | 5.5         |
| SP         | 41         | 6.2         | 22.8         | 25.9        | 808         | 4.1         |
| <b>All</b> | <b>411</b> | <b>69.5</b> | <b>228.3</b> | <b>24.5</b> | <b>7904</b> | <b>39.7</b> |

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 113: plots on the 300 ft grid (orange = QA flag)

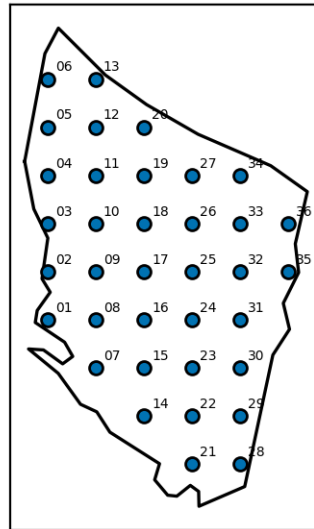


Figure 1. Plot locations with QA status.

**QA findings:** no issues found; unit accepted

Table 4. Plot list

| Plot   | Trees in | BA sq ft/ac | Slope % | Status |
|--------|----------|-------------|---------|--------|
| 113-01 | 13       | 260         | 26      | ok     |
| 113-02 | 14       | 280         | 27      | ok     |
| 113-03 | 11       | 220         | 27      | ok     |
| 113-04 | 11       | 220         | 24      | ok     |
| 113-05 | 13       | 260         | 28      | ok     |
| 113-06 | 13       | 260         | 31      | ok     |
| 113-07 | 9        | 180         | 21      | ok     |
| 113-08 | 13       | 260         | 16      | ok     |
| 113-09 | 11       | 220         | 35      | ok     |
| 113-10 | 12       | 240         | 21      | ok     |
| 113-11 | 12       | 240         | 33      | ok     |
| 113-12 | 11       | 220         | 20      | ok     |
| 113-13 | 11       | 220         | 23      | ok     |
| 113-14 | 12       | 240         | 23      | ok     |
| 113-15 | 11       | 220         | 33      | ok     |
| 113-16 | 13       | 260         | 25      | ok     |
| 113-17 | 11       | 220         | 22      | ok     |
| 113-18 | 12       | 240         | 11      | ok     |
| 113-19 | 9        | 180         | 32      | ok     |
| 113-20 | 10       | 200         | 19      | ok     |
| 113-21 | 12       | 240         | 35      | ok     |
| 113-22 | 10       | 200         | 30      | ok     |
| 113-23 | 12       | 240         | 28      | ok     |
| 113-24 | 12       | 240         | 27      | ok     |
| 113-25 | 10       | 200         | 25      | ok     |
| 113-26 | 10       | 200         | 27      | ok     |
| 113-27 | 10       | 200         | 32      | ok     |
| 113-28 | 10       | 200         | 21      | ok     |
| 113-29 | 12       | 240         | 17      | ok     |
| 113-30 | 12       | 240         | 10      | ok     |
| 113-31 | 9        | 180         | 10      | ok     |
| 113-32 | 11       | 220         | 9       | ok     |
| 113-33 | 12       | 240         | 15      | ok     |
| 113-34 | 15       | 300         | 17      | ok     |
| 113-35 | 11       | 220         | 13      | ok     |
| 113-36 | 11       | 220         | 11      | ok     |